

1 Document structure

L^AT_EX document can be divided into three main parts;

1. Document class
2. Preamble
3. Document body

1.1 Document class

This is the first line of L^AT_EX document, and defines the type (or class) of document. Some of the common classes include:

- **article** → to write articles/short reports. Top level header is **section** and from there it goes to **subsection**, **subsubsection** and so on.
- **book** → for books. Supports **part** and **chapter** levels for writing. And if I recall correctly, has different margins for odd/even pages.
- **beamer** → for creating presentations.

In addition to defining the type of document, the command `\documentclass` also supports optional arguments. Some of the options are:

- **papersize** → a4, a5, letter etc.
- **fontsize** → font size are defined in terms of *pt*. Default is 10pt.
- **orientation** → landscape/portrait. Default is portrait.

An example of document class is:

Listing 1: Document class

```
1 \documentclass[a4paper, 10pt]{article}
```

1.2 Preamble

This section defines all the packages that are required for the document, settings that apply to whole document, user based commands (or macros) as well as document metadata. An example of preamble is presented in Listing 2:

Listing 2: Defining document preamble

```
1 % packages
2 \usepackage[colorlinks=true]{hyperref}
3 \usepackage[utf8]{inputenc}
4 \usepackage[T1]{fontenc}
5 \usepackage[left=2.54cm, right=2.54cm, top=2.54cm, bottom=2.54cm]{
  geometry}
6
7 % settings for document
8 \setlength{\parindent}{0pt}
9
10 % metadata
11 \author{names}
12 \title{text}
13 \date{text}
```

1.3 Document body

This is the part where actual text (text that you want to put in document) goes. The contents are enclosed by `\begin{document}` and `\end{document}` commands. Inside document body, one can include different environments like figures, tables, equations and so on.

Some common typesetting tips:

- `\textbf{Bold text}` → **Bold text**
- `\textit{Italic text}` → *Italic text*
- `\underline{Underline text}` → Underline text
- `A\textsubscript{subscript}` → $A_{\text{subscript}}$
- `A\textsuperscript{superscript}` → $A^{\text{superscript}}$
- `\textcolor{Blue}{blue text}` → blue text
- `\newline` or `\\` → for line break.
- `\newpage` → forcing page break.

2 Equation

Writing equation in $\text{\LaTeX}$ is very simple. Equations can be written in line using `$` operator, or they can be written inside math environment. See listing 3 for example

Listing 3: Writing equations in $\text{\LaTeX}$

```
1 Equations can be written inline e.g. Relativity states  $E=mc^2$ 
2
3 Equation can also be written in display mode using the \texttt{equation
  } environment:
4 \begin{equation}
5     E = mc^2
6 \end{equation}
```

Equations can be written inline e.g. Relativity states $E = mc^2$

Equation can also be written in display mode using the `equation` environment:

$$E = mc^2 \tag{1}$$

$\text{\LaTeX}$ also supports writing subequations (listing 4). Writing greek/latin characters is also quite intuitive, normally the character is preceded by a backslash (`\`) e.g. `\rho` → ρ . Note that for writing Greek/Latin characters within text requires the same syntax as inline equations, i.e. enclosing it between `$` operator. Few more things to note in listing 4;

- the command `\label{key}` is used to create an anchor for cross-referencing the object at some other point in document.
- Math (or equation environment) normally writes everything in *italics*. `\mathrm{}` formats the math symbols to normal (roman) font.
- Spacing is ignored in math mode. To add space between characters or operators use `\` followed by space.

- Simple operators like addition and subtraction can be written as keyboard characters (+ or -). For multiplication use `\times` or `\cdot` $\rightarrow \times$ or $\cdot$ respectively. For division, use `\frac{num}{den}` $\rightarrow \frac{num}{den}$

Listing 4: Writing subequations, fractions, greek/latin characters

```

1 \begin{subequations}
2 \begin{gather}
3 \mathrm{uc}_{\{f\}}|_{\{z = 0\}} = \mathrm{uc}_{\{f\}}|_{\{z = 0^+\}} - D_{\{ax\}}\frac{\partial c_f}{\partial z}\frac{\partial}{\partial z}\Big|_{\{z = 0^+\}} \label{eq:BC-inlet} \\
4 \mathrm{frac}{\partial c_f(z = L, t)}{\partial z} = 0 \label{eq:BC-outlet}
5 \end{gather}
6 \label{eq:BoundaryConditions}
7 \end{subequations}

```

$$\mathrm{uc}_f|_{z=0} = \mathrm{uc}_f|_{z=0^+} - D_{ax}\frac{\partial c_f}{\partial z}\Big|_{z=0^+} \quad (2a)$$

$$\frac{\partial c_f(z = L, t)}{\partial z} = 0 \quad (2b)$$

3 Figures

Figures in $\text{\LaTeX}$ are included using `figure` environment. The supported types are `.pdf`, `.eps`, `.png`, `.jpeg`. General syntax for including figures is given in listing 5. Important to include packages `graphicx` and `float` in document preamble. The latter is required for changing figure position.

Listing 5: Including figures in $\text{\LaTeX} 2_{\epsilon}$

```

1 \begin{figure}[H]
2 \centering
3 \includegraphics[width=0.6\textwidth]{learning_curve.png}
4 \caption[Comparison between different tools for writing text]{
5   Comparison between different tools for writing text. \LaTeX{} is a
6   bit difficult to start with, but becomes easier to write with time.
7   Markdown is also a nice way of writing documents. See \href{https://
8   quarto.org/docs/authoring/markdown-basics.html}{Quarto} for a good
9   markdown option.}
10 \label{fig:demo_figure}
11 \end{figure}

```

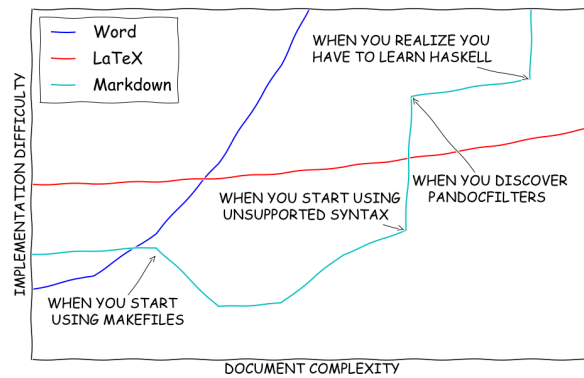


Figure 1: Comparison between different tools for writing text. $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ is a bit difficult to start with, but becomes easier to write with time. Markdown is also a nice way of writing documents. See [Quarto](#) for a good markdown option.